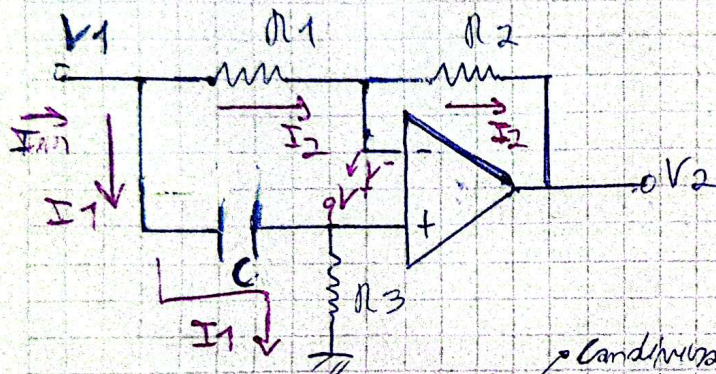


Ejercicio 4)

- a) Filtro Paso Bajas o Retardador de Fase, queremos $H(s) = \frac{V_2(s)}{V_1(s)}$ tanto el módulo, la fase y el diagrama de Bode y cero.
- Como $\frac{R_2}{R_1} = 1$ y $\frac{R_1}{R_3} = 5$



- Asumo filtro ideal
- $V^+ = V^-$ por equipotencialidad
- $Y_C(s) = sC$, $Y_R(s) = \frac{1}{R}$

• $V^+ = V^- = V_1 \cdot \frac{Y_C(s)}{Y_C(s) + Y_{R3}(s)}$ Condición de tensión
 $I_{in}(s) = I_1(s) + I_2(s)$

• $I_1(s) = (V_1(s) - V^+) \cdot Y_C(s) = (V^+ - 0V) \cdot Y_{R3}(s) \Rightarrow V_1(s) \cdot Y_C(s) - V^+ \cdot Y_C(s) = V^+ \cdot Y_{R3}(s)$
 $\Rightarrow V_1(s) \cdot Y_C(s) = V^+ \cdot Y_C(s) + V^+ \cdot Y_{R3}(s) \Rightarrow V_1(s) \cdot Y_C(s) = V^+ (Y_C(s) + Y_{R3}(s))$
 $\Rightarrow \frac{V_1(s) \cdot Y_C(s)}{Y_C(s) + Y_{R3}(s)} = V^+(s) \rightarrow$ por igualdad de conductancias.

• $I_2(s) = (V_1(s) - V^-) \cdot Y_{R1}(s) = (V^- - V_2(s)) \cdot Y_{R2}(s)$
 $\Rightarrow I_2(s) = V_1(s) \cdot Y_{R1}(s) - V^- \cdot Y_{R1}(s) = V^- \cdot Y_{R2}(s) - V_2(s) \cdot Y_{R2}(s)$
 $\Rightarrow V_1(s) \cdot Y_{R1}(s) + V_2(s) \cdot Y_{R2}(s) = V^- \cdot Y_{R2}(s) + V^- \cdot Y_{R1}(s)$
 $V_1(s) \cdot Y_{R1}(s) + V_2(s) \cdot Y_{R2}(s) = V^- (Y_{R2}(s) + Y_{R1}(s))$
 $V_1(s) \cdot Y_{R1}(s) + V_2(s) \cdot Y_{R2}(s) = V_1(s) \cdot \frac{Y_C(s)}{Y_C(s) + Y_{R3}(s)} \cdot (Y_{R2}(s) + Y_{R1}(s))$

$$V_2(s) \cdot Y_{R2}(s) = V_1(s) \cdot \left(\frac{Y_C(s) \cdot Y_{R2}(s)}{Y_C(s) + Y_{R3}(s)} + \frac{Y_C(s) \cdot Y_{R1}(s)}{Y_C(s) + Y_{R3}(s)} \right) \cdot \frac{V_1(s)}{V_1(s)}$$

$$\Rightarrow \frac{V_2(s)}{V_1(s)} = \frac{Y_C(s) \cdot Y_{R2}(s)}{Y_C(s) + Y_{R3}(s)} \cdot \frac{1}{Y_{R2}(s)} + \frac{Y_C(s) \cdot Y_{R1}(s)}{Y_C(s) + Y_{R3}(s)} \cdot \frac{1}{Y_{R2}(s)} - \frac{Y_{R1}(s)}{Y_{R2}(s)}$$

$$\frac{V_2(s)}{V_1(s)} = \frac{V_c(s)}{V_c(s) + V_{n2}(s)} \cdot \left(1 + \frac{V_{n1}(s)}{V_{n2}(s)} \right) - \frac{V_{n1}(s)}{V_{n2}(s)}$$

$$\frac{V_2(s)}{V_1(s)} = \frac{g_c}{g_c + \frac{1}{n_3}} \cdot \left(1 + \frac{1/n_1}{1/n_2} \right) - \frac{1/n_1}{1/n_2}$$

$$\frac{V_2(s)}{V_1(s)} = \frac{g_c n_3 \left(1 + \frac{n_2}{n_1} \right) - \frac{n_2}{n_1}}{\left(g_c + \frac{1}{n_3} \right) n_3}$$

$$\frac{V_2(s)}{V_1(s)} = \frac{g_c n_3}{g_c n_3 + 1} \left(1 + \frac{n_2}{n_1} \right) - \frac{n_2}{n_1} \quad (A)$$

$$H(s) = \frac{g_c n_3 n_1 \left(1 + \frac{n_2}{n_1} \right) - (g_c n_3 + 1) n_2}{(g_c n_3 + 1) \cdot n_1}$$

$$H(s) = \frac{g_c n_3 n_1 + g_c n_3 n_2 - g_c n_3 n_2 - n_2}{g_c n_3 n_1 + n_1}$$

$$H(s) = \frac{g_c n_3 n_1 - n_2}{g_c n_3 n_1 + n_1} \rightarrow \text{Transfencia, expresión general} \quad (B)$$

Partiendo de (A) $\Rightarrow H(s) = \frac{g_c n_3 (1+1) - 1}{g_c n_3 + 1}$

$$H(s) = \frac{2 \cdot g_c n_3}{g_c n_3 + 1} - 1$$

$$H(s) = \frac{2 g_c n_3 - (g_c n_3 + 1)}{g_c n_3 + 1}$$

$$H(s) = \frac{2 \cdot g_c n_3 - g_c n_3 - 1}{g_c n_3 + 1}$$

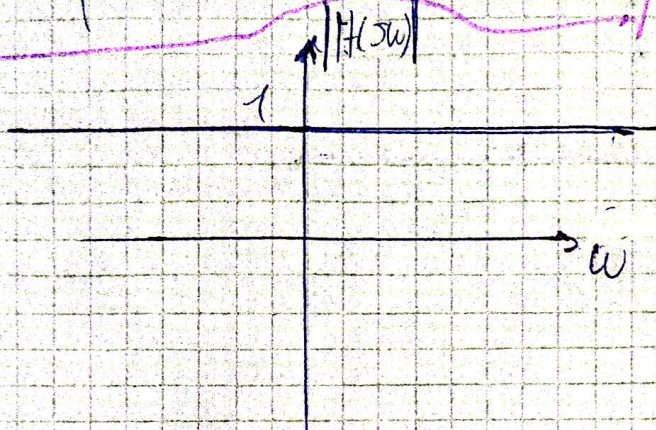
$$H(s) = \frac{g_c n_3 - 1}{g_c n_3 + 1}$$

$$\text{con } \frac{n_2}{n_1} = 1$$

Busco el módulo

$$s = j\omega \Rightarrow H(s) = \frac{j\omega \cdot c \cdot \alpha_3 - 1}{j\omega \cdot c \cdot \alpha_3 + 1} \Rightarrow |H(j\omega)| = \frac{\sqrt{(-1)^2 + (\omega \cdot c \cdot \alpha_3)^2}}{\sqrt{1^2 + (\omega \cdot c \cdot \alpha_3)^2}}$$

$\Rightarrow |H(j\omega)| = 1 \rightarrow$ Es un pasapasa $\rightarrow$ módulo



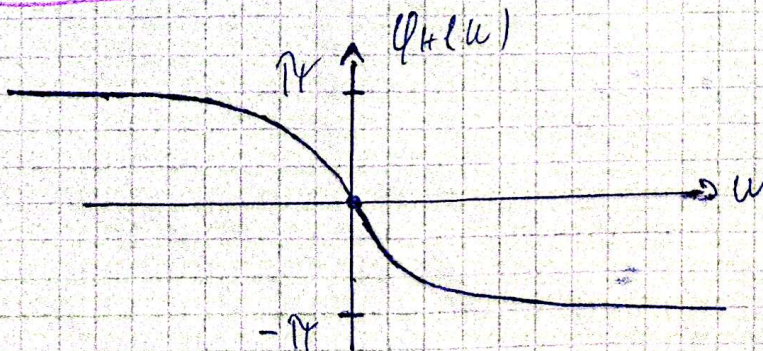
Busco la fase:

$$\varphi_H(\omega) = \arctg\left(\frac{\text{Im}\{N_{H(s)}\}}{\text{Re}\{N_{H(s)}\}}\right) - \arctg\left(\frac{\text{Im}\{D_{H(s)}\}}{\text{Re}\{D_{H(s)}\}}\right)$$

$$\varphi_H(\omega) = \arctg\left(\frac{\omega \cdot c \cdot \alpha_3}{-1}\right) - \arctg\left(\frac{\omega \cdot c \cdot \alpha_3}{1}\right)$$

$$\varphi_H(\omega) = -\arctg(\omega \cdot c \cdot \alpha_3) - \arctg(\omega \cdot c \cdot \alpha_3)$$

$\varphi_H(\omega) = -2 \cdot \arctg(\omega \cdot c \cdot \alpha_3) \rightarrow$ Fase



$\bullet \omega = 0 \Rightarrow \varphi_H(0) = 0$

$\bullet \omega \rightarrow +\infty \Rightarrow \varphi_H(\omega) \rightarrow -\pi$

$\bullet \omega \rightarrow -\infty \Rightarrow \varphi_H(\omega) \rightarrow \pi$

Diagrama de polos de zeros:

$$H(s) = \frac{s \cdot 10^3 - 1}{s \cdot 10^3 + 1} \quad \frac{-1/10^3}{1/10^3} = \frac{s - 1/10^3}{s + 1/10^3} = \frac{s - 1000}{s + 1000}$$

• Numerador da "zeros" $\rightarrow s_0 = \frac{1}{10^3}$

• Denominador do "polos" $\rightarrow s_x = -\frac{1}{10^3}$

• $10^3 = 10^{-3}$, $\frac{1}{10^3} = 1000$

